

# Enhanced Multi-Data-Center Power–Computing Coordinated Scheduling Model

## 1 System Description

Consider a multi-period power–computing coordinated system over a scheduling horizon  $\mathcal{T}$ . The electrical layer is based on a distribution network, which contains thermal units, wind power units, and photovoltaic (PV) units. Multiple geographically distributed data centers are connected to different buses of the distribution system. Each data center contains multiple computing nodes that can be turned on or off according to workload demand.

Two types of workloads are considered:

- **Interactive workload:** latency-sensitive requests generated by different source regions. Each request flow must be assigned to one feasible data center in each time period.
- **Batch workload:** delay-tolerant jobs with release times and deadlines. Batch jobs can be processed and migrated among data centers over time.

The system operates in a grid-connected mode. Bus 1 is treated as the root/slack bus through which the distribution network exchanges power with the upstream grid. The objective is to jointly optimize thermal generation dispatch, renewable power utilization, power imported from the upstream grid, data center activation, node-level workload allocation, internal data-center operation, and workload migration while satisfying the operating constraints of both the power network and the computing system.

## 2 Objective Function

The objective is to minimize the total operating cost:

$$\min C^{\text{th}} + C^{\text{grid}} + C^{\text{sw}} + C^{\text{mig},I} + C^{\text{mig},B} + C^{\text{cur}} + C^{\text{late}}. \quad (1)$$

### 2.1 Thermal Generation Cost

$$C^{\text{th}} = \sum_{t \in \mathcal{T}} \sum_{g \in \mathcal{G}^{\text{th}}} \left( a_g (P_{g,t}^{\text{th}})^2 + b_g P_{g,t}^{\text{th}} + c_g u_{g,t} \right). \quad (2)$$

### 2.2 Upstream Grid Purchase Cost

$$C^{\text{grid}} = \sum_{t \in \mathcal{T}} \pi_t^{\text{grid}} P_t^{\text{grid}}. \quad (3)$$

### 2.3 Node Switching Cost

$$C^{\text{sw}} = \sum_{d \in \mathcal{D}} \sum_{n \in \mathcal{N}_d} \pi_{d,n}^{\text{sw}} \eta_{d,n,1} + \sum_{t \in \mathcal{T} \setminus \{1\}} \sum_{d \in \mathcal{D}} \sum_{n \in \mathcal{N}_d} \pi_{d,n}^{\text{sw}} \eta_{d,n,t}. \quad (4)$$

## 2.4 Interactive Workload Switching Cost

$$C^{\text{mig},I} = \sum_{t \in \mathcal{T} \setminus \{1\}} \sum_{r \in \mathcal{R}} \sum_{\substack{d, d' \in \mathcal{D} \\ d \neq d'}} \pi_r^{\text{mig},I} m_{r,d,d',t}^I. \quad (5)$$

## 2.5 Batch Job Migration Cost

$$C^{\text{mig},B} = \sum_{t \in \mathcal{T} \setminus \{1\}} \sum_{j \in \mathcal{J}} \sum_{\substack{d, d' \in \mathcal{D} \\ d \neq d'}} \pi_j^{\text{mig},B} m_{j,d,d',t}^B. \quad (6)$$

## 2.6 Renewable Curtailment Cost

$$C^{\text{cur}} = \sum_{t \in \mathcal{T}} \left( \sum_{g \in \mathcal{G}^w} \pi_g^{\text{cur},w} P_{g,t}^{\text{cur},w} + \sum_{g \in \mathcal{G}^{\text{pv}}} \pi_g^{\text{cur},\text{pv}} P_{g,t}^{\text{cur},\text{pv}} \right). \quad (7)$$

## 2.7 Batch Deadline Penalty

$$C^{\text{late}} = \sum_{j \in \mathcal{J}} \pi_j^{\text{late}} (1 - \delta_j). \quad (8)$$

# 3 Constraints

## 3.1 Upstream Grid Exchange Constraints

$$\begin{aligned} 0 \leq P_t^{\text{grid}} &\leq \overline{P}^{\text{grid}}, & \forall t \in \mathcal{T}, \\ \underline{Q}^{\text{grid}} \leq Q_t^{\text{grid}} &\leq \overline{Q}^{\text{grid}}, & \forall t \in \mathcal{T}. \end{aligned} \quad (9)$$

## 3.2 Simplified Thermal Unit Constraints

### Active power output limits

$$\underline{P}_g^{\text{th}} u_{g,t} \leq P_{g,t}^{\text{th}} \leq \overline{P}_g^{\text{th}} u_{g,t}, \quad \forall g \in \mathcal{G}^{\text{th}}, \forall t \in \mathcal{T}. \quad (11)$$

### Reactive power output limits

$$\underline{Q}_g^{\text{th}} u_{g,t} \leq Q_{g,t}^{\text{th}} \leq \overline{Q}_g^{\text{th}} u_{g,t}, \quad \forall g \in \mathcal{G}^{\text{th}}, \forall t \in \mathcal{T}. \quad (12)$$

### Ramping constraints For the first period:

$$-RD_g \leq P_{g,1}^{\text{th}} - P_{g,0}^{\text{ini}} \leq RU_g, \quad \forall g \in \mathcal{G}^{\text{th}}. \quad (13)$$

For the remaining periods:

$$-RD_g \leq P_{g,t}^{\text{th}} - P_{g,t-1}^{\text{th}} \leq RU_g, \quad \forall g \in \mathcal{G}^{\text{th}}, \forall t \in \mathcal{T} \setminus \{1\}. \quad (14)$$

### 3.3 Renewable Generation Constraints

#### Wind power constraints

$$0 \leq P_{g,t}^w \leq \bar{P}_{g,t}^w, \quad \forall g \in \mathcal{G}^w, \forall t \in \mathcal{T}, \quad (15)$$

$$P_{g,t}^w + P_{g,t}^{\text{cur},w} = \bar{P}_{g,t}^w, \quad \forall g \in \mathcal{G}^w, \forall t \in \mathcal{T}, \quad (16)$$

$$\underline{Q}_g^w \leq Q_{g,t}^w \leq \bar{Q}_g^w, \quad \forall g \in \mathcal{G}^w, \forall t \in \mathcal{T}. \quad (17)$$

#### PV power constraints

$$0 \leq P_{g,t}^{\text{pv}} \leq \bar{P}_{g,t}^{\text{pv}}, \quad \forall g \in \mathcal{G}^{\text{pv}}, \forall t \in \mathcal{T}, \quad (18)$$

$$P_{g,t}^{\text{pv}} + P_{g,t}^{\text{cur,pv}} = \bar{P}_{g,t}^{\text{pv}}, \quad \forall g \in \mathcal{G}^{\text{pv}}, \forall t \in \mathcal{T}, \quad (19)$$

$$\underline{Q}_g^{\text{pv}} \leq Q_{g,t}^{\text{pv}} \leq \bar{Q}_g^{\text{pv}}, \quad \forall g \in \mathcal{G}^{\text{pv}}, \forall t \in \mathcal{T}. \quad (20)$$

### 3.4 Interactive Workload Constraints

#### Unique assignment over latency-feasible data centers

$$\sum_{d \in \mathcal{D}_r^{\text{lat}}} x_{r,d,t}^I = 1, \quad \forall r \in \mathcal{R}, \forall t \in \mathcal{T}. \quad (21)$$

#### Infeasible assignments are forbidden

$$x_{r,d,t}^I = 0, \quad \forall r \in \mathcal{R}, \forall d \in \mathcal{D} \setminus \mathcal{D}_r^{\text{lat}}, \forall t \in \mathcal{T}. \quad (22)$$

#### Interactive workload received by each data center

$$L_{d,t}^I = \sum_{r \in \mathcal{R}} \phi_r^I \lambda_{r,t} x_{r,d,t}^I, \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}. \quad (23)$$

#### Interactive assignment switching

$$m_{r,d,d',t}^I \geq x_{r,d,t-1}^I + x_{r,d',t}^I - 1, \quad \forall r \in \mathcal{R}, \forall d \neq d', \forall t \in \mathcal{T} \setminus \{1\}. \quad (24)$$

### 3.5 Batch Workload Constraints

#### Location uniqueness within the valid time window

$$\sum_{d \in \mathcal{D}} z_{j,d,t} = 1, \quad \forall j \in \mathcal{J}, \forall t \in \mathcal{T}_j. \quad (25)$$

#### No location before release time or after deadline

$$z_{j,d,t} = 0, \quad \forall j \in \mathcal{J}, \forall d \in \mathcal{D}, \forall t \in \mathcal{T} \setminus \mathcal{T}_j. \quad (26)$$

#### Execution only at the assigned data center

$$x_{j,d,n,t}^B \leq z_{j,d,t}, \quad \forall j \in \mathcal{J}, \forall d \in \mathcal{D}, \forall n \in \mathcal{N}_d, \forall t \in \mathcal{T}. \quad (27)$$

### Single-node execution per time period

$$\sum_{d \in \mathcal{D}} \sum_{n \in \mathcal{N}_d} x_{j,d,n,t}^B \leq 1, \quad \forall j \in \mathcal{J}, \forall t \in \mathcal{T}. \quad (28)$$

### Processing amount constraints

$$0 \leq f_{j,d,n,t} \leq C_{d,n}^{\text{cpu}} x_{j,d,n,t}^B, \quad \forall j \in \mathcal{J}, \forall d \in \mathcal{D}, \forall n \in \mathcal{N}_d, \forall t \in \mathcal{T}, \quad (29)$$

$$x_{j,d,n,t}^B = 0, \quad f_{j,d,n,t} = 0, \quad \forall j \in \mathcal{J}, \forall d \in \mathcal{D}, \forall n \in \mathcal{N}_d, \forall t \in \mathcal{T} \setminus \mathcal{T}_j, \quad (30)$$

$$\sum_{t \in \mathcal{T}_j} \sum_{d \in \mathcal{D}} \sum_{n \in \mathcal{N}_d} f_{j,d,n,t} \geq W_j \delta_j, \quad \forall j \in \mathcal{J}. \quad (31)$$

### Batch migration constraint

$$m_{j,d,d',t}^B \geq z_{j,d,t-1} + z_{j,d',t} - 1, \quad \forall j \in \mathcal{J}, \forall d \neq d', \forall t \in \mathcal{T} \setminus \{1\}. \quad (32)$$

## 3.6 Data Center Activation and Capacity Constraints

### Node activation implies data center activation

$$s_{d,n,t} \leq y_{d,t}, \quad \forall d \in \mathcal{D}, \forall n \in \mathcal{N}_d, \forall t \in \mathcal{T}. \quad (33)$$

### Activation consistency

$$\sum_{n \in \mathcal{N}_d} s_{d,n,t} \leq |\mathcal{N}_d| y_{d,t}, \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}, \quad (34)$$

$$y_{d,t} \leq \sum_{n \in \mathcal{N}_d} s_{d,n,t}, \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}. \quad (35)$$

### Batch jobs can only be executed on active nodes

$$x_{j,d,n,t}^B \leq s_{d,n,t}, \quad \forall j \in \mathcal{J}, \forall d \in \mathcal{D}, \forall n \in \mathcal{N}_d, \forall t \in \mathcal{T}. \quad (36)$$

### Joint computing capacity constraint

$$L_{d,t}^I + \sum_{j \in \mathcal{J}} \sum_{n \in \mathcal{N}_d} f_{j,d,n,t} \leq \sum_{n \in \mathcal{N}_d} C_{d,n}^{\text{cpu}} s_{d,n,t}, \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}. \quad (37)$$

## 3.7 Complex Internal Data Center Operation Model

To preserve the original outer-layer structure while introducing a more sophisticated internal data-center model, we explicitly distinguish minimum-QoS service nodes, redundant service nodes, and peak-power service nodes. This leads to a more strongly coupled internal operation model for IT power, cooling power, and indoor temperature.

First, define the total batch workload processed in data center  $d$  at time  $t$  as

$$F_{d,t}^B = \sum_{j \in \mathcal{J}} \sum_{n \in \mathcal{N}_d} f_{j,d,n,t}, \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}. \quad (38a)$$

Introduce the following additional binary variables:

- $a_{d,n,t}^I$ : node  $n$  is assigned to minimum-QoS interactive service;
- $a_{d,n,t}^B$ : node  $n$  is assigned to minimum-QoS batch service;
- $r_{d,n,t}^I$ : node  $n$  is assigned as redundant interactive-service resource;
- $r_{d,n,t}^B$ : node  $n$  is assigned as redundant batch-service resource;
- $w_{d,n,t}^I$ : node  $n$  is forced to operate at peak mode for interactive service;
- $w_{d,n,t}^B$ : node  $n$  is forced to operate at peak mode for batch service.

**Node-role consistency** A node can take at most one service role in each time period:

$$a_{d,n,t}^I + a_{d,n,t}^B + r_{d,n,t}^I + r_{d,n,t}^B \leq s_{d,n,t}, \quad \forall d \in \mathcal{D}, \forall n \in \mathcal{N}_d, \forall t \in \mathcal{T}. \quad (38b)$$

Peak-mode operation is only allowed for nodes assigned to the corresponding service:

$$w_{d,n,t}^I \leq a_{d,n,t}^I + r_{d,n,t}^I, \quad \forall d \in \mathcal{D}, \forall n \in \mathcal{N}_d, \forall t \in \mathcal{T}, \quad (38c)$$

$$w_{d,n,t}^B \leq a_{d,n,t}^B + r_{d,n,t}^B, \quad \forall d \in \mathcal{D}, \forall n \in \mathcal{N}_d, \forall t \in \mathcal{T}. \quad (38d)$$

**Aggregated service-node counts** Define the numbers of nodes in different service states:

$$m_{d,t}^{IM} = \sum_{n \in \mathcal{N}_d} a_{d,n,t}^I, \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}, \quad (38e)$$

$$m_{d,t}^{BM} = \sum_{n \in \mathcal{N}_d} a_{d,n,t}^B, \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}, \quad (38f)$$

$$m_{d,t}^{IR} = \sum_{n \in \mathcal{N}_d} r_{d,n,t}^I, \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}, \quad (38g)$$

$$m_{d,t}^{BR} = \sum_{n \in \mathcal{N}_d} r_{d,n,t}^B, \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}, \quad (38h)$$

$$m_{d,t}^{IP} = \sum_{n \in \mathcal{N}_d} w_{d,n,t}^I, \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}, \quad (38i)$$

$$m_{d,t}^{BP} = \sum_{n \in \mathcal{N}_d} w_{d,n,t}^B, \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}. \quad (38j)$$

**Minimum service constraints** The minimum number of service nodes must be sufficient for interactive and batch workloads:

$$L_{d,t}^I \leq \left( \mu_d - \frac{1}{v_d} \right) m_{d,t}^{IM}, \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}, \quad (38k)$$

$$F_{d,t}^B \leq \mu_d m_{d,t}^{BM}, \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}. \quad (38l)$$

**Role-feasibility constraints** The total number of service and redundant nodes cannot exceed the active nodes:

$$m_{d,t}^{IM} + m_{d,t}^{BM} + m_{d,t}^{IR} + m_{d,t}^{BR} \leq \sum_{n \in \mathcal{N}_d} s_{d,n,t}, \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}. \quad (38m)$$

The number of peak-mode nodes cannot exceed the number of nodes participating in the corresponding service:

$$m_{d,t}^{IP} \leq m_{d,t}^{IM} + m_{d,t}^{IR}, \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}, \quad (38n)$$

$$m_{d,t}^{BP} \leq m_{d,t}^{BM} + m_{d,t}^{BR}, \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}. \quad (38o)$$

**IT power consumption model** The minimum-QoS IT power is modeled as

$$P_{d,t}^{ITM} = P_d^{NET} \frac{m_{d,t}^{IM} + m_{d,t}^{BM}}{|\mathcal{N}_d|} + e_d^I (m_{d,t}^{IM} + m_{d,t}^{BM}) + \frac{e_d^P - e_d^I}{\mu_d} (L_{d,t}^I + F_{d,t}^B), \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}. \quad (39a)$$

The redundant/high-QoS IT power is modeled as

$$P_{d,t}^{ITH} = P_d^{NET} \frac{m_{d,t}^{IR} + m_{d,t}^{BR}}{|\mathcal{N}_d|} + e_d^I (m_{d,t}^{IR} + m_{d,t}^{BR}) + \frac{e_d^P - e_d^I}{\mu_d} \left[ m_{d,t}^{IP} \left( \mu_d - \frac{L_{d,t}^I}{m_{d,t}^{IM} + m_{d,t}^{IR} + \varepsilon} \right) + m_{d,t}^{BP} \left( \mu_d - \frac{F_{d,t}^B}{m_{d,t}^{BM} + m_{d,t}^{BR} + \varepsilon} \right) \right], \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}, \quad (39b)$$

where  $\varepsilon > 0$  is a sufficiently small positive constant used to avoid division by zero.

The total IT power is

$$P_{d,t}^{IT} = P_{d,t}^{ITM} + P_{d,t}^{ITH}, \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}. \quad (39c)$$

The IT power cannot exceed the total peak power of active nodes:

$$P_{d,t}^{IT} \leq \sum_{n \in \mathcal{N}_d} P_{d,n}^{\text{peak}} s_{d,n,t}, \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}. \quad (39d)$$

**Cooling and thermal dynamics** The cooling electric power is modeled as a linear function of cooling thermal power:

$$P_{d,t}^C = a_d^C h_{d,t} + b_d^C, \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}. \quad (39e)$$

Let

$$\kappa_d = \exp \left( -\frac{\Delta t}{R_d C_d} \right).$$

Then the indoor temperature evolution is

$$\theta_{d,t} = \kappa_d \theta_{d,t-1} + (1 - \kappa_d) T_{d,t}^{\text{out}} + R_d (1 - \kappa_d) (P_{d,t}^{IT} + P_d^O - h_{d,t}), \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}. \quad (39f)$$

The indoor temperature and cooling-power limits are

$$\underline{\theta}_d \leq \theta_{d,t} \leq \bar{\theta}_d, \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}, \quad (40a)$$

$$0 \leq h_{d,t} \leq \bar{h}_d, \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}, \quad (40b)$$

$$\underline{P}_d^C \leq P_{d,t}^C \leq \bar{P}_d^C, \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}. \quad (40c)$$

Finally, the total active and reactive powers of data center  $d$  are

$$P_{d,t}^{DC} = P_{d,t}^{IT} + P_{d,t}^C + P_d^O, \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}, \quad (40d)$$

$$Q_{d,t}^{DC} = P_{d,t}^{DC} \tan(\arccos(pf_d)), \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}. \quad (40e)$$

### 3.8 Node Switching Cost Linearization

For the first period

$$\eta_{d,n,1} \geq s_{d,n,1} - s_{d,n,0}^{\text{ini}}, \quad \forall d \in \mathcal{D}, \forall n \in \mathcal{N}_d, \quad (41)$$

$$\eta_{d,n,1} \geq s_{d,n,0}^{\text{ini}} - s_{d,n,1}, \quad \forall d \in \mathcal{D}, \forall n \in \mathcal{N}_d. \quad (42)$$

For the remaining periods

$$\eta_{d,n,t} \geq s_{d,n,t} - s_{d,n,t-1}, \quad \forall d \in \mathcal{D}, \forall n \in \mathcal{N}_d, \forall t \in \mathcal{T} \setminus \{1\}, \quad (43)$$

$$\eta_{d,n,t} \geq s_{d,n,t-1} - s_{d,n,t}, \quad \forall d \in \mathcal{D}, \forall n \in \mathcal{N}_d, \forall t \in \mathcal{T} \setminus \{1\}, \quad (44)$$

$$\eta_{d,n,t} \geq 0, \quad \forall d \in \mathcal{D}, \forall n \in \mathcal{N}_d, \forall t \in \mathcal{T}. \quad (45)$$

### 3.9 Distribution Network Constraints

The total active and reactive demands at bus  $i$  are defined as

$$P_{i,t}^{\text{load,tot}} = P_{i,t}^L + \sum_{d \in \mathcal{D}: b(d)=i} P_{d,t}^{DC}, \quad \forall i \in \mathcal{B}, \forall t \in \mathcal{T}, \quad (46)$$

$$Q_{i,t}^{\text{load,tot}} = Q_{i,t}^L + \sum_{d \in \mathcal{D}: b(d)=i} Q_{d,t}^{DC}, \quad \forall i \in \mathcal{B}, \forall t \in \mathcal{T}. \quad (47)$$

The total active generation at bus 1 is defined as

$$P_{1,t}^{\text{gen}} = P_t^{\text{grid}} + \sum_{g \in \mathcal{G}_1^{\text{th}}} P_{g,t}^{\text{th}} + \sum_{g \in \mathcal{G}_1^{\text{w}}} P_{g,t}^{\text{w}} + \sum_{g \in \mathcal{G}_1^{\text{pv}}} P_{g,t}^{\text{pv}}, \quad \forall t \in \mathcal{T}. \quad (48)$$

The total reactive generation at bus 1 is defined as

$$Q_{1,t}^{\text{gen}} = Q_t^{\text{grid}} + \sum_{g \in \mathcal{G}_1^{\text{th}}} Q_{g,t}^{\text{th}} + \sum_{g \in \mathcal{G}_1^{\text{w}}} Q_{g,t}^{\text{w}} + \sum_{g \in \mathcal{G}_1^{\text{pv}}} Q_{g,t}^{\text{pv}}, \quad \forall t \in \mathcal{T}. \quad (49)$$

For buses  $i \in \mathcal{B} \setminus \{1\}$ , the total active and reactive generation are

$$P_{i,t}^{\text{gen}} = \sum_{g \in \mathcal{G}_i^{\text{th}}} P_{g,t}^{\text{th}} + \sum_{g \in \mathcal{G}_i^{\text{w}}} P_{g,t}^{\text{w}} + \sum_{g \in \mathcal{G}_i^{\text{pv}}} P_{g,t}^{\text{pv}}, \quad \forall i \in \mathcal{B} \setminus \{1\}, \forall t \in \mathcal{T}, \quad (50)$$

$$Q_{i,t}^{\text{gen}} = \sum_{g \in \mathcal{G}_i^{\text{th}}} Q_{g,t}^{\text{th}} + \sum_{g \in \mathcal{G}_i^{\text{w}}} Q_{g,t}^{\text{w}} + \sum_{g \in \mathcal{G}_i^{\text{pv}}} Q_{g,t}^{\text{pv}}, \quad \forall i \in \mathcal{B} \setminus \{1\}, \forall t \in \mathcal{T}. \quad (51)$$

A branch-flow (DistFlow) model is adopted.

$$P_{ij,t} = \sum_{k:(j,k) \in \mathcal{L}} P_{jk,t} + P_{j,t}^{\text{load,tot}} - P_{j,t}^{\text{gen}} + r_{ij}\ell_{ij,t}, \quad \forall (i,j) \in \mathcal{L}, \forall t \in \mathcal{T}, \quad (52)$$

$$Q_{ij,t} = \sum_{k:(j,k) \in \mathcal{L}} Q_{jk,t} + Q_{j,t}^{\text{load,tot}} - Q_{j,t}^{\text{gen}} + x_{ij}\ell_{ij,t}, \quad \forall (i,j) \in \mathcal{L}, \forall t \in \mathcal{T}, \quad (53)$$

$$V_{j,t} = V_{i,t} - 2(r_{ij}P_{ij,t} + x_{ij}Q_{ij,t}) + (r_{ij}^2 + x_{ij}^2)\ell_{ij,t}, \quad \forall (i,j) \in \mathcal{L}, \forall t \in \mathcal{T}, \quad (54)$$

$$P_{ij,t}^2 + Q_{ij,t}^2 \leq V_{i,t}\ell_{ij,t}, \quad \forall (i,j) \in \mathcal{L}, \forall t \in \mathcal{T}, \quad (55)$$

$$\underline{V}_i \leq V_{i,t} \leq \bar{V}_i, \quad \forall i \in \mathcal{B}, \forall t \in \mathcal{T}, \quad (56)$$

$$P_{ij,t}^2 + Q_{ij,t}^2 \leq \bar{S}_{ij}^2, \quad \forall (i,j) \in \mathcal{L}, \forall t \in \mathcal{T}, \quad (57)$$

$$V_{1,t} = V^{\text{slack}}, \quad \forall t \in \mathcal{T}. \quad (58)$$

## 4 Variable Domains

$$u_{g,t} \in \{0, 1\}, \quad \forall g \in \mathcal{G}^{\text{th}}, \forall t \in \mathcal{T}, \quad (59)$$

$$s_{d,n,t}, y_{d,t} \in \{0, 1\}, \quad \forall d \in \mathcal{D}, \forall n \in \mathcal{N}_d, \forall t \in \mathcal{T}, \quad (60)$$

$$a_{d,n,t}^I, a_{d,n,t}^B, r_{d,n,t}^I, r_{d,n,t}^B, w_{d,n,t}^I, w_{d,n,t}^B \in \{0, 1\}, \quad \forall d \in \mathcal{D}, \forall n \in \mathcal{N}_d, \forall t \in \mathcal{T}, \quad (60a)$$

$$x_{r,d,t}^I \in \{0, 1\}, \quad \forall r \in \mathcal{R}, \forall d \in \mathcal{D}, \forall t \in \mathcal{T}, \quad (61)$$

$$m_{r,d,d',t}^I \in \{0, 1\}, \quad \forall r \in \mathcal{R}, \forall d \neq d', \forall t \in \mathcal{T} \setminus \{1\}, \quad (62)$$

$$z_{j,d,t}, x_{j,d,n,t}^B, m_{j,d,d',t}^B, \delta_j \in \{0, 1\}, \quad \forall j \in \mathcal{J}, \forall d, d' \in \mathcal{D}, \forall n \in \mathcal{N}_d, \forall t \in \mathcal{T}. \quad (63)$$

The remaining variables are continuous, with nonnegativity imposed where needed.

## 5 Compact Formulation

The overall power-computing coordinated scheduling problem can be summarized as follows:

$$\min C^{\text{th}} + C^{\text{grid}} + C^{\text{sw}} + C^{\text{mig},I} + C^{\text{mig},B} + C^{\text{cur}} + C^{\text{late}} \quad \text{s.t. constraints (9)–(37), (38a)–(40e), (41)–(63).} \quad (64)$$

## 6 Model Characteristics

The proposed model has the following characteristics:

- It explicitly couples the power system and the computing system through the data-center internal power model.
- It operates in a grid-connected mode by introducing upstream-grid power injection at the root bus.



- It distinguishes interactive and batch workloads by using different routing, service, and migration constraints.
- It preserves the original inter-data-center interaction structure, including interactive workload re-routing and batch-job migration among geographically distributed data centers.
- It enriches the data-center internal model by introducing minimum-QoS service nodes, redundant service nodes, and peak-power service nodes, leading to a more complex IT-power representation.
- The overall model generally belongs to a mixed-integer nonlinear programming framework due to the strongly coupled internal data-center constraints.

## 7 Remarks

1. The reactive power capability of thermal, wind, and PV units is approximated by independent box constraints for tractability.
2. If reactive power optimization is not needed, all reactive-power variables and constraints can be removed to obtain a simpler active-power-only model.
3. If all batch jobs must be completed,  $\delta_j$  can be fixed to 1 for all  $j \in \mathcal{J}$ .
4. If parallel execution of batch jobs is allowed, constraint (28) can be removed.
5. If interactive workload splitting among multiple data centers is needed,  $x_{r,d,t}^I$  can be relaxed from binary variables to continuous assignment ratios.
6. If a more tractable approximation is desired, the nonlinear terms in (39b) can be linearized or approximated by piecewise-linear methods.

**Additional internal data-center parameters** The enhanced internal model uses the following additional parameters:

- $\mu_d$ : nominal processing rate of a service node in data center  $d$ ;
- $v_d$ : tolerated response-delay parameter for interactive service in data center  $d$ ;
- $P_d^{NET}$ : network/switching power component associated with active service nodes in data center  $d$ ;
- $e_d^I$ : idle power of an active service node in data center  $d$ ;
- $e_d^P$ : peak power of an active service node in data center  $d$ ;
- $P_{d,n}^{\text{peak}}$ : peak IT power of node  $n$  in data center  $d$ ;
- $P_d^O$ : other non-IT and non-cooling power consumption in data center  $d$ ;
- $a_d^C, b_d^C$ : cooling-power coefficients;
- $R_d, C_d$ : equivalent thermal resistance and thermal capacitance of data center  $d$ ;
- $T_{d,t}^{\text{out}}$ : outdoor temperature at data center  $d$  and time  $t$ ;

- $\underline{\theta}_d, \bar{\theta}_d$ : lower and upper indoor-temperature limits;
- $\bar{h}_d$ : upper bound of cooling thermal power;
- $\underline{P}_d^C, \bar{P}_d^C$ : lower and upper bounds of cooling electric power;
- $\varepsilon$ : a sufficiently small positive constant used to regularize the denominator in (39b).